

The Doh Problem

Given a poly time DTM M , $\underbrace{\text{decision versus search}}_{\text{polynomially related to each other}}$ version of problem $\underbrace{\text{does there exist/find}}_{\text{polynomially related to each other}}$ input y that causes $M(y)$ to accept.

Doh is in NP

Given solution y , check if $M(y)$ accepts.

Doh is in NP-Hard

Suppose have access to oracle L for Doh. Given any NP problem, by definition, there must exist a poly time DTM $M'(x,y)$ for verifying solution y . Ask oracle for L , whether there exists a y that causes M' to accept after hardwiring x in description of M' .

So, $\text{Doh} \in \text{NPC}$.

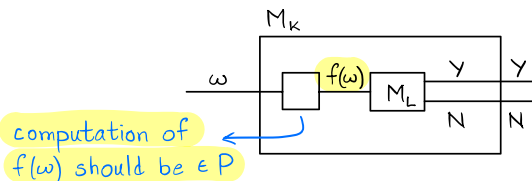
Proving a problem is NPC

Now that we know $\text{Doh} \in \text{NPC}$, we can prove a new problem L is NPC by

1. Proving $L \in \text{NP}$
2. Polytime reduce a known NPC language K to L , i.e., $K \leq_p L$.
(Hence proving $L \in \text{NP-Hard}$).

Polytime Reductions

$K \leq_p L$
poly time mapping reduction



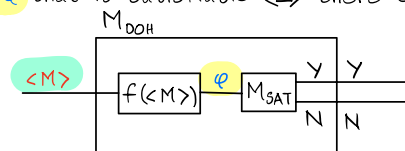
SAT Given an arbitrary boolean formula is there an assignment to its variables that causes the formula to evaluate to True, e.g.,

$$\varphi = (\bar{x} \wedge y) \vee (x \wedge \bar{z})$$

$$\text{SAT} = \{ \varphi \mid \varphi \text{ is a satisfiable boolean formula} \}$$

Theorem $\text{SAT} \in \text{NPC}$.

1. $\text{SAT} \in \text{NP}$ (Given proposed solution just plugin & evaluate)
2. $\text{Doh} \leq_p \text{SAT}$
 Given an arbitrary polytime DTM M , need to create a formula φ that is satisfiable $\Leftrightarrow$ there exists a y that is accepted by M .



Note that given TM M & its input $y = y_1 y_2 \dots y_r$, future of M is determined. We can construct a table which represents entire history of M 's computation, which runs in polytime $p(r)$. Let $p(r) = T$.

each row has configuration info of M at time t

Time Step	M's state	Tape Contents
T	q_{acc}	11 0 ... 110 ... 1
T-1	.	010 ... 1 0 ... 1
.	.	010 ... 0 0 ... 1
.	.	.
4	q_c	.
3	q_b	.
2	q_a	y_1 y₂ y_3 ... y_r
1	q_{init}	y₁ y_2 y_3 ... y_r

initial head location ← i

The table has T rows & each row is $O(T)$, so overall size is poly in input size. The initial configuration has y as input that shows up as variables in the formula we construct.

Variables

Tape Content 1. For all i, t let $x_{t,i} = 1 \Leftrightarrow$ at time step t , i^{th} cell on tape contains 1.

Head location 2. Let $p_{t,i} = 1 \Leftrightarrow$ at time step t , head points to i^{th} cell.

M's internal state 3. Let $s_{t,k} = 1 \Leftrightarrow$ at time step t , M is in state k .

Clauses for φ

1. $x_{1,i} = y_i$ for $1 \leq i \leq r$ &
 $x_{1,i} = 0$ for $i > r$ } Input on tape at $t=1$

2. $p_{t,i} = 0 \Rightarrow (x_{t+1,i} = x_{t,i})$ cell content is unchanged if head doesn't point to it

3. $(s_{t,k} = 1) \wedge (p_{t,i} = 1) \wedge (x_{t,i} = 1) \Rightarrow s_{t+1,k'} = 1$

if M is in state k at time t
 &
 head points to i^{th} cell at time t
 &
 i^{th} cell on tape contains 1 at time t
 then M is in state k' at time $t+1$

Transition function of M
 Similarly, for all $l \neq k'$, $s_{t+1,l} = 0$

Finally $s_{T,acc} = 1$. (End in accept state)

We can combine all these clauses to get a φ that is satisfiable
 $\Leftrightarrow$

there exists a y that is accepted by M .

Special Cases of SAT

A boolean formula φ is in **Conjunctive Normal Form (CNF)** if it is the **AND** of **ORs** of literals, where a literal is a variable or its negation, e.g.,

$$\varphi = \bigwedge_{i=1}^n \underbrace{(x_{k,i} \vee \overline{x}_{l,i})}_{\text{clause}}$$

(Note: $x_{k,i}$ and $\overline{x}_{l,i}$ are literals, and the OR is over literals)

A CNF formula is **2CNF** if every clause involves at most 2 literals.

$$2SAT = \{ \varphi \mid \varphi \text{ is in 2CNF \& is satisfiable} \}$$

Exercise: $2SAT \in P$.

However, $3SAT \in NPC$.

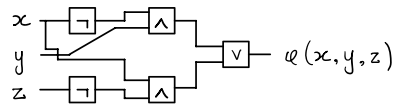
We will show $SAT \leq_p \text{CircuitSAT} \leq_p 3SAT$.

$SAT \leq_p \text{CircuitSAT}$

We can convert every boolean formula φ into an equivalent circuit C in linear time.

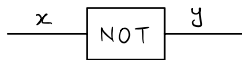
e.g.,

$$\varphi(x, y, z) = (\overline{x} \wedge y) \vee (x \wedge \overline{z})$$



Now, to show $\text{CircuitSAT} \leq_p 3SAT$, we note each gate has at most three wires, two inputs & one output. We define these as variables for each gate & enforce them so that the input & output values satisfy the gate condition.

e.g.,



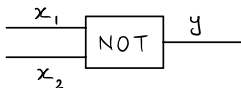
We want if x is true then y is false
& if x is false then y is true

x	y	Is $y = \text{NOT}(x)$
0	0	0
0	1	1
1	0	1
1	1	0

$$(p \Rightarrow q) = (\neg p \vee q)$$

$$\begin{aligned} x &\Leftrightarrow \overline{y} \\ &= (x \Rightarrow \overline{y}) \wedge (\overline{y} \Rightarrow x) \\ &= (\overline{x} \vee \overline{y}) \wedge (x \vee y) \end{aligned}$$

Similarly, $y = \text{OR}(x_1, x_2)$



x_1	x_2	y	Is $y = \text{OR}(x_1, x_2)$
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

OR

$$(\neg x_1 \vee \neg x_2 \vee y) \wedge (\neg x_1 \vee x_2 \vee y) \wedge (x_1 \vee \neg x_2 \vee y) \wedge (x_1 \vee x_2 \vee \neg y)$$

Exercise: Perform same construction for AND.

3Color

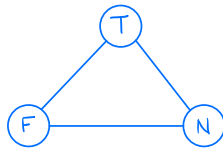
Given a graph G , is it possible to color each vertex $\{\text{Red, Green, Blue}\}$ so that no adjacent vertex has same color.

$3\text{Color} \in \text{NPC}$. We show $3\text{SAT} \leq_p 3\text{Color}$

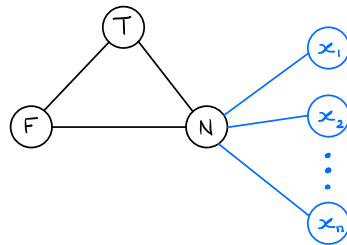
Given a 3CNF formula φ as input $= C_1 \wedge C_2 \wedge \dots \wedge C_n$, where C_i is a single clause.

Map $\{\text{Red, Green, Blue}\}$ to labels $\{\text{True, False, N}\}$.

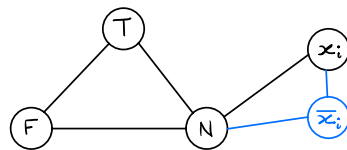
Construct following triangle & one vertex for each variable in φ .



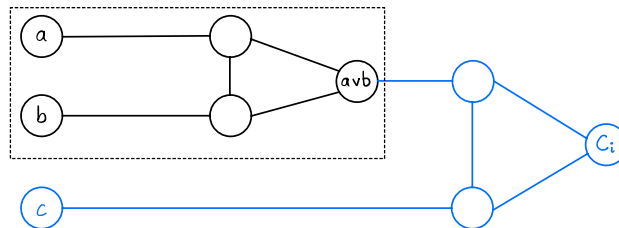
We create one vertex for each variable x_i & connect them to vertex $\textcircled{N}$ to force their color to only T or F.



If a variable x_i has its negation in the formula then we create a new vertex for $\bar{x}_i$ & make them adjacent.



For a clause $C_i = a \vee b \vee c$ we construct the OR gadget.



Finally, since each C_i needs to be True for φ to be satisfiable, we connect each C_i to N & F.

